



BrainTT: Distinguishing Glioma Recurrence from Radiation Necrosis on Post-Treatment Brain MRI

<https://github.com/Emoliere-s-hajimi/LZU-INFO442-BrainTT>.

<https://braintt.vercel.app>

Team 14

Leader: Yutong Wang (ID: 320230942461)

Member: Zijin Wu(ID:320230942571), Xiaopeng Fan(ID:320230941801),
Ye Wang(ID:320230942541), Yunfei Shang(ID:320230942271)

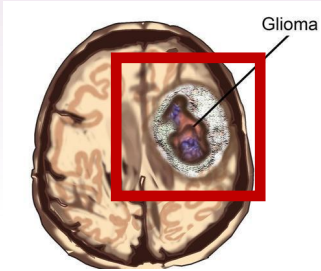
CONTENTS

- 1 Motivation and Values
- 2 Data Acquisition and Processing
- 3 Research Questions and Methods
- 4 Experiments and Results
- 5 Team and Roles

The clinical scenario

Gliomas

A deadly brain cancer



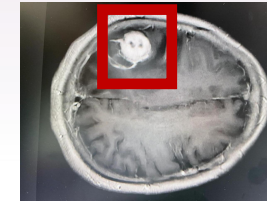
Treatment & Follow up

Surgery and regular brain MRI scans



New Lesion

Bright spots often appear on followup scan



2 possible answers

A. The tumor is back
or
B. The radiation has injured healthy tissue

On a routine MRI, these two very different things often look nearly the same, even for the best doctors. That ambiguity is the question our project tries to answer.

The tumor is back

RECURRENCE

Needs aggressive second-line chemotherapy or re-irradiation. **MUST started as early as possible. If mis-call as necrosis, patient's cancer go UNTREATED.**

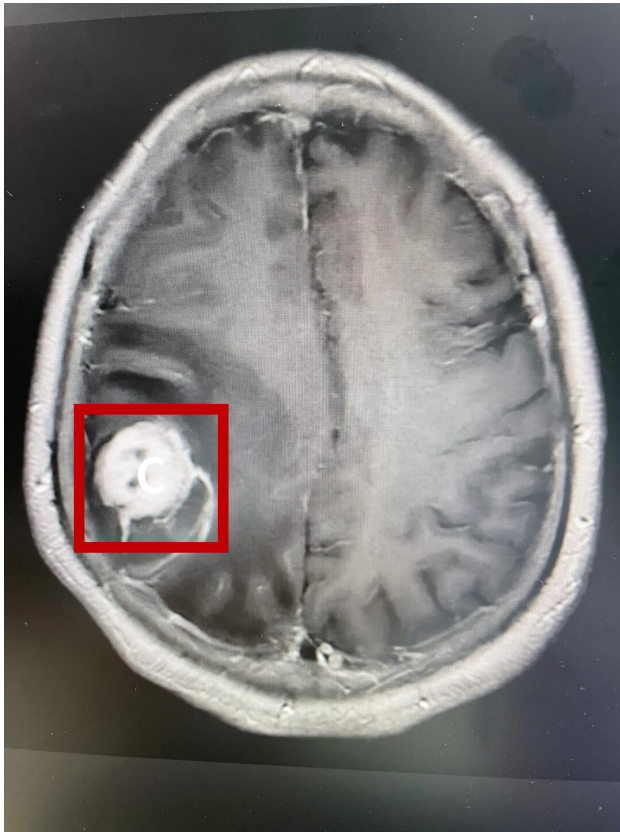
The radiation has injured healthy tissue

RADIATION NECROSIS

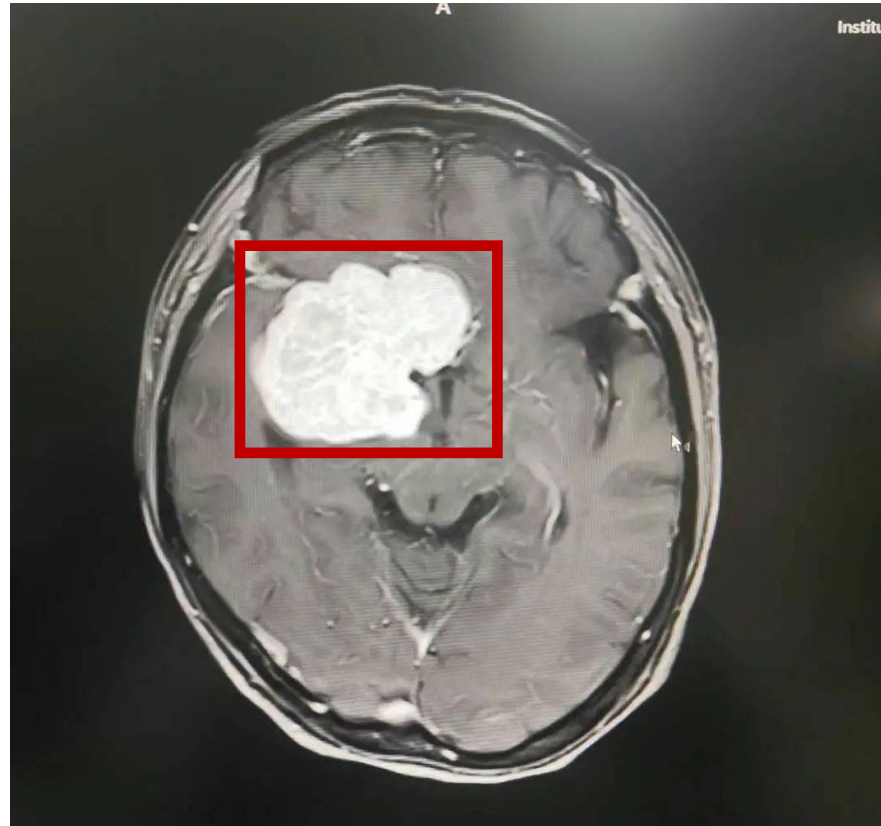
Needs conservative anti-inflammatory care. **If mis-call as necrosis, patient receives harmful chemo / re-radiation for no benefit.**

The clinical scenario

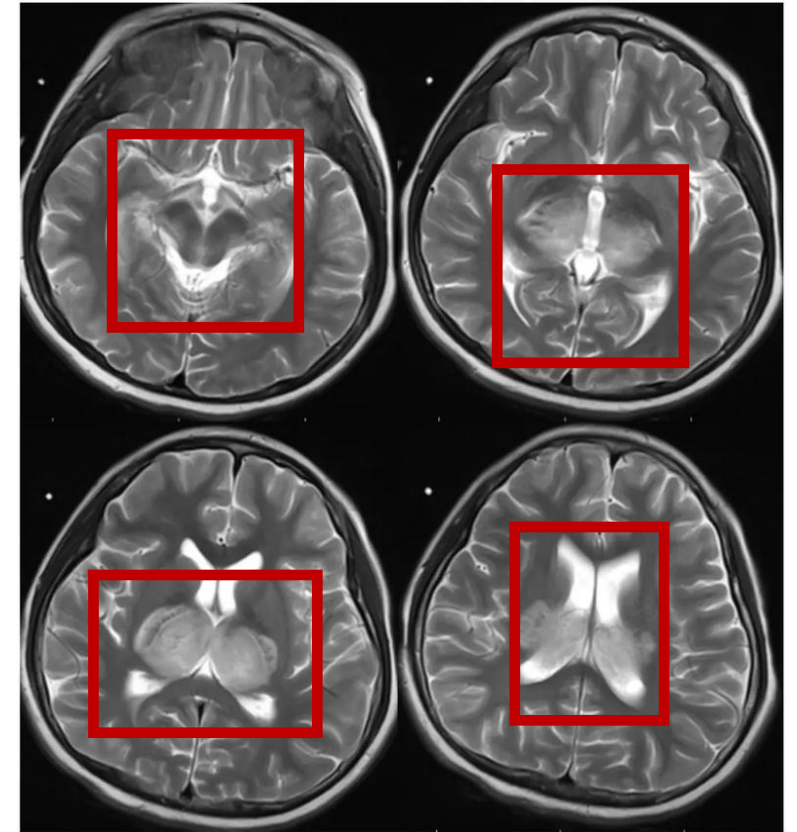
Can you try to distinguish Glioma Recurrence from Radiation Necrosis?



Necrosis



Recurrence



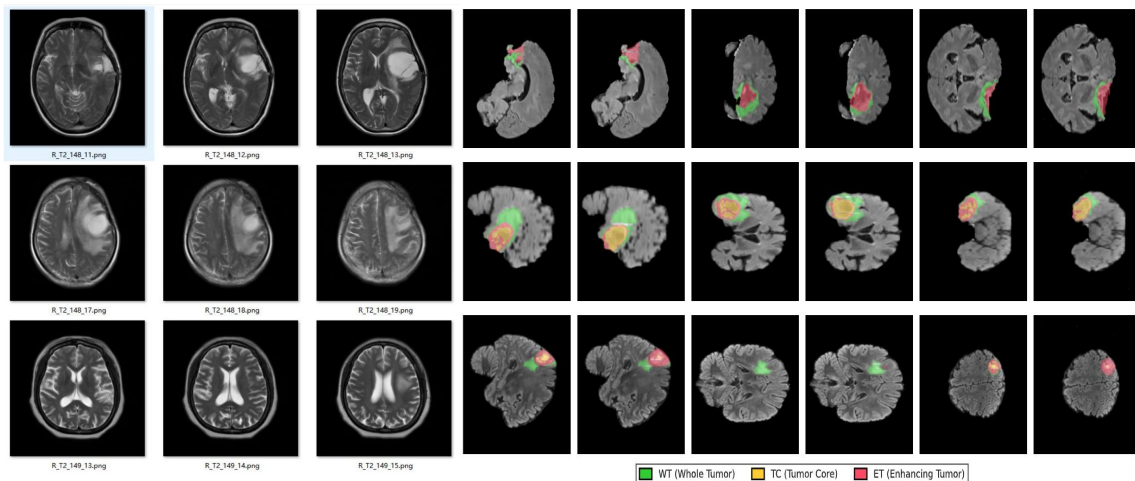
Recurrence

Why Solving It Matters, and Why It's Still Open

- If a model could give a reliable non-invasive answer here, the value lands in several places at once:
- Patients are spared brain surgery they did not need, offering new insights for future scientific research

This problem has been studied. But none of the existing tools has entered routine clinical use. There are structural reasons:

From Data



WT (Whole Tumor) TC (Tumor Core) ET (Enhancing Tumor)

- Most public datasets are made from patients BEFORE treatment, ours' are AFTER treatment
- Who label the data? Cost too much time and money !!!

From Methodology

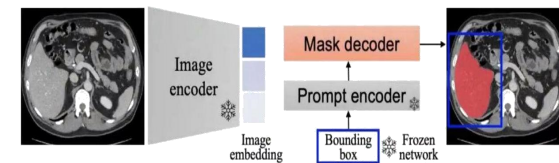
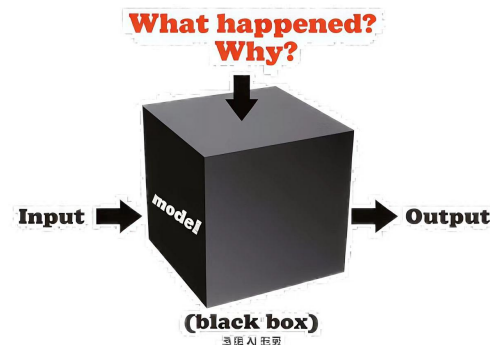


Figure 3: MedSAM: Fine-tuning Segment Anything Model for medical image segmentation by freezing the image encoder and prompt encoder and only fine-tuning the mask decoder.



CONTENTS

1

Motivation and Values

2

Data Acquisition and Processing

3

Research Questions and Methods

4

Experiments and Results

5

Team and Roles

Data Provenance: A Horizontal Collaboration

- **Beijing Tiantan Hospital**, one of the strongest hospitals in neuroscience in the world
- Institute of Software from **Chinese Academy of Sciences & Lanzhou University**

Tiantan Hospital

Zhulin An, CAS

Zhongfeng Kang, LZU

Development and validation of a d... MRI.pdf 1.4M

胶质瘤复发与坏死研究_阶段1.pdf 10.0M

天坛医院_胶质瘤坏死与复发_20210619.pdf 1.7M

好的

OK

通过网盘分享的文件

链接: h...
pwd=1...

Files shared via cloud storage: 2 files, including Cai Xiaohong's handover materials.
Link: h...
P...

这是之前的数据和代码, 你也看下吧

Here are the previous data and code. Take a look at them as well.

好的, 谢谢老师

Okay, thank you, teacher.

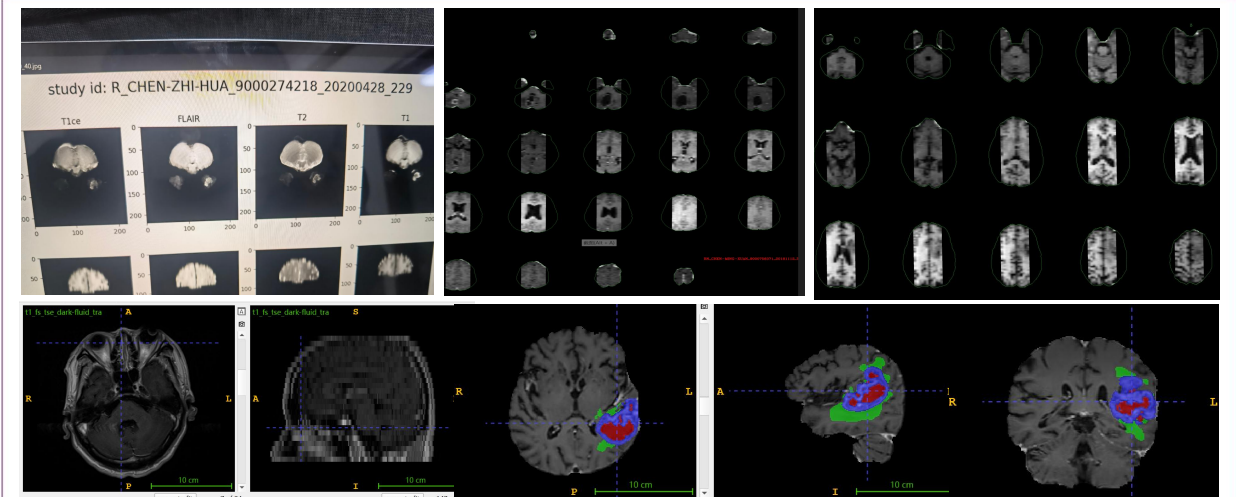
之前一些材料, 你看先, 别客气

Take a look at some of the previous materials first. Feel free to do so.

老师我大致看了发现这些数据格式跟我之前做的基本上是一样的

Teacher, after taking a cursory glance, I noticed that the data formats used in these tasks are largely the same as those I used previously.

Our team has been granted access to the dataset



Tiantan hospital
collect clinical
origins and labels

Prof. Zhulin An
conducted the
first cleaning

Prof. Kang
improved the
dataset

Through this collaboration, a private dataset containing 234 patients with 2500+ cases was born

Cohort Characteristics

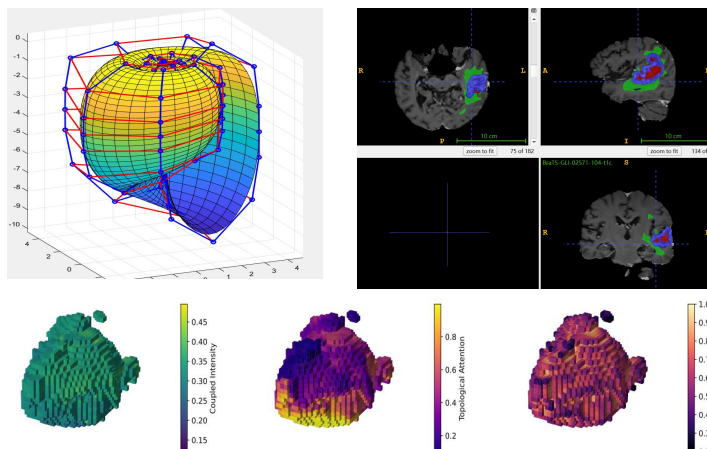
- 234 patients , tracing from Jan 2012 to Dec 2022, containing 1-4 MRI sequences per case
- Much larger than existing dataset like BraTS, BrainTumor

Large Scale



- ❑ 31.9GB (just t1 Necrosis data)
- ❑ 234 patients, 2537 patient cases, totally 89.7GB .tar

Complex Features



- ❑ 3D images, 6.6MB per cases
- ❑ Extreme sample ratio, tumor area usually < 1% of the brain

Totally Ethical

1_0.dcm	12_11.dcm	11_10.dcm
2_1.dcm	13_12.dcm	12_11.dcm
3_2.dcm	14_13.dcm	13_12.dcm
4_3.dcm	15_14.dcm	14_13.dcm
5_4.dcm	16_15.dcm	15_14.dcm
6_5.dcm	17_16.dcm	16_15.dcm
7_6.dcm	18_17.dcm	17_16.dcm
8_7.dcm	19_18.dcm	18_17.dcm
9_8.dcm	20_19.dcm	19_18.dcm
10_9.dcm	21_20.dcm	20_19.dcm
11_10.dcm	22_21.dcm	21_20.dcm
	23_22.dcm	22_21.dcm
	24_23.dcm	23_22.dcm

- ❑ All private patients' information are removed, replaced with numbers

Large scale dataset

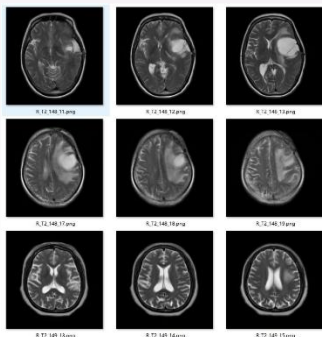
Multimodal 3D MRI

No ethical issues

Data Pipeline

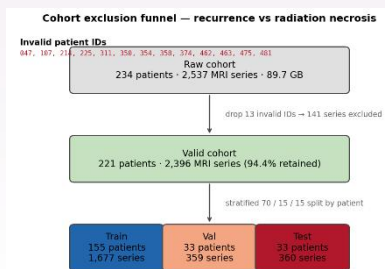
Raw cohort

- N / R / RN case folders
- T1, T1ce, T2, FLAIR or missing
- Private clinical cohort



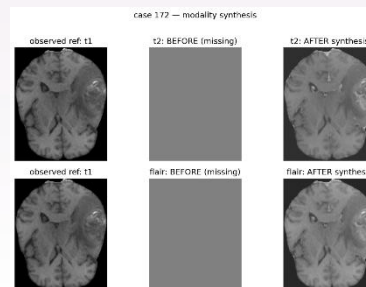
Cleaning

- Resolve class labels
- Prefer revised N labels
- Drop invalid / unusable cases



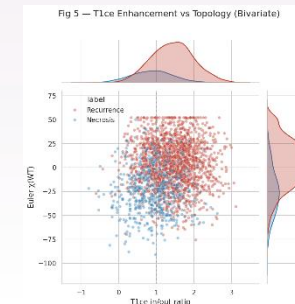
Preprocessing

- Resample to 1 mm grid
- Foreground z-score
- Synthesize missing channels



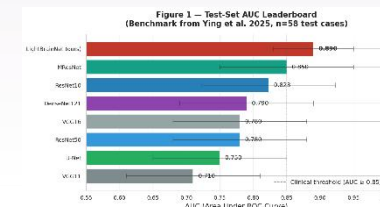
EDA

- T1ce enhancement ratio
- Euler χ topology
- Class imbalance
- Modality ranking



BrainTTNet

- Modality prior
- Topology prior
- Anatomy prior



Key design bridge

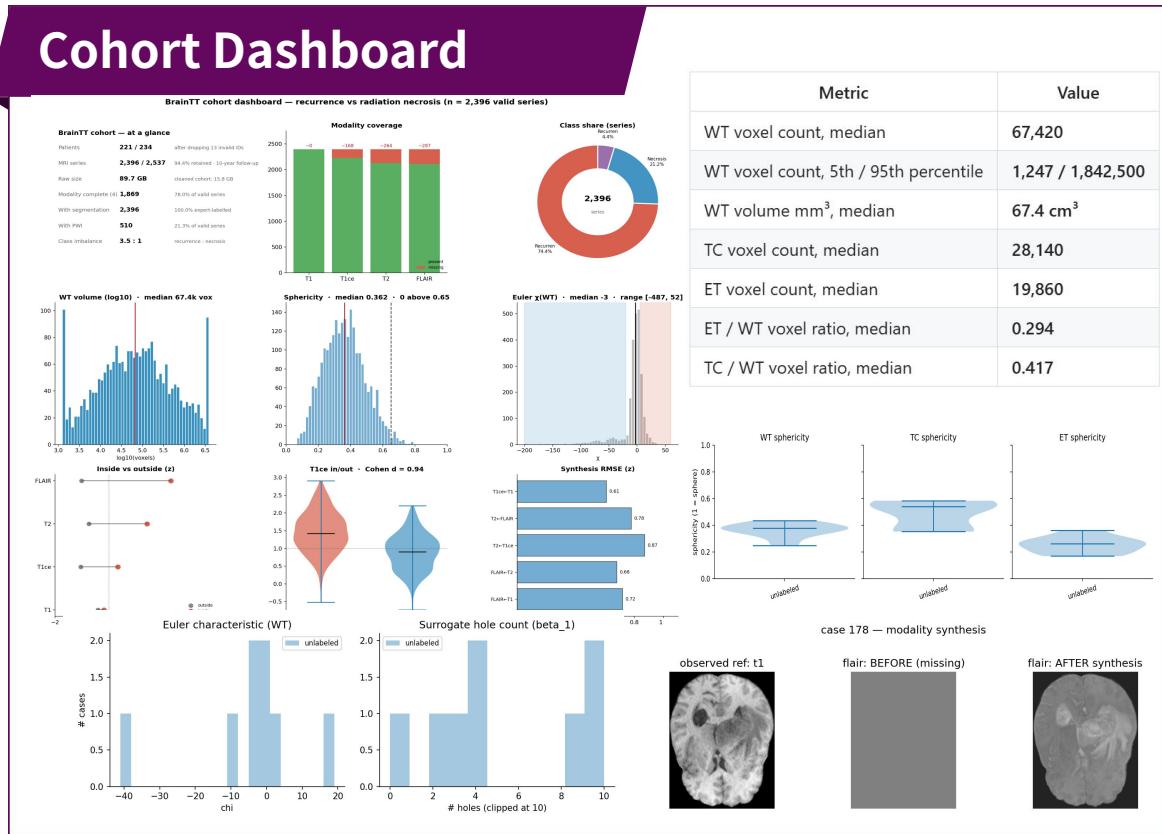
T1ce enhancement and **Euler χ topology** discovered in EDA were converted into BrainTTNet's modality and topology priors.

Basic EDA and Insights

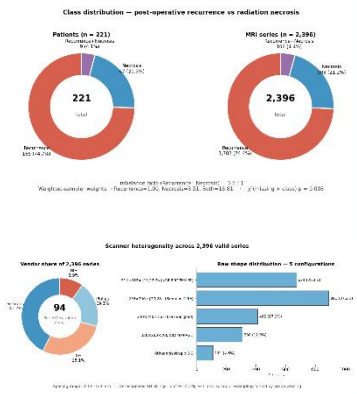
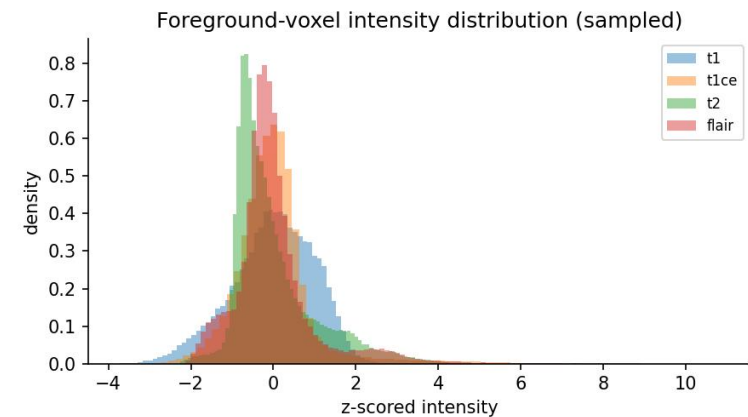
EDA Focus

EDA profiles a cleaned **10-year** cohort with **221** valid patients, **2,396** MRI series, complete labels, and real-world acquisition variability.

Cohort Dashboard



Distribution & Heterogeneity



Data Quality

Model Risk

- Valid cohort: **221** patients, **2,396** MRI series, **100%** segmentation coverage.
- Imbalance, missing modalities, and scanner heterogeneity** require robust model design.

3.5:1 class imbalance, 22% missing-modality rate, 4-vendor scanner heterogeneity, and long-tailed lesion morphology.

Prior Mining

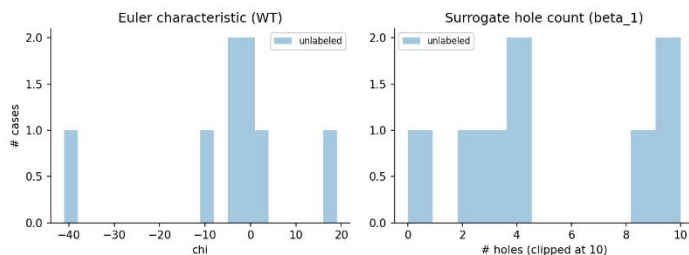
Cohort-level imaging patterns are converted into explicit priors, guiding modality fusion, topology-aware representation, and anatomy-constrained segmentation.

- **22%** missing modalities
- Strong **FLAIR/T2** contrast
- **T1ce** separates classes

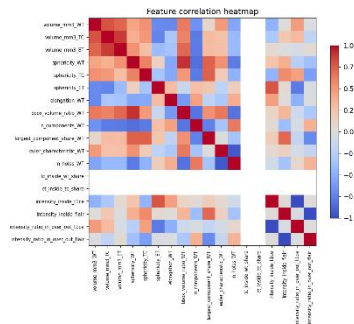
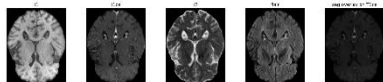
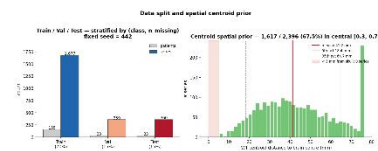
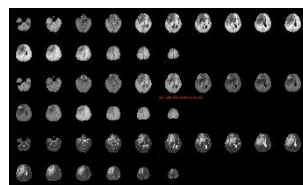
- **4-order** volume range
- Sphericity = **0.362**
- **Non-spherical** tumors

- **38%** multifocal lesions
- Compact vs cavitated topology
- **Non-intensity** discrimination

MRI Modality Examples



Topology Signatures from WT Masks



Feature Correlation & Auxiliary Priors

Modality Evidence

Modality Coupling Prior

- Mask-aware fusion
- Contrast-guided attention
- Missing-channel compensation

Morphology Evidence

Topology-Shape Prior

- Non-spherical patterns
- Multi-scale gradients
- Euler χ regularization

Spatial Evidence

Anatomy-Spatial Prior

- Center-biased masks
- Boundary suppression
- Spatial localization prior

Mined Cohort Patterns

Translate Into

Prior-Guided Model Modules

CONTENTS

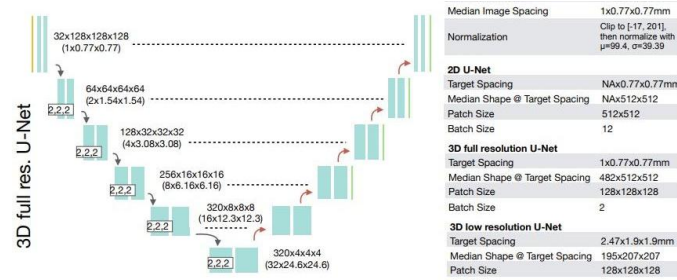
- 1 Motivation and Values
- 2 Data Acquisition and Processing
- 3 Research Questions and Methods
- 4 Experiments and Results
- 5 Team and Roles

Research Questions

3 types of audiences

- For methods reviewers, for the statisticians, and for the radiologists
- Simultaneously satisfying all three is the precondition for clinical usage

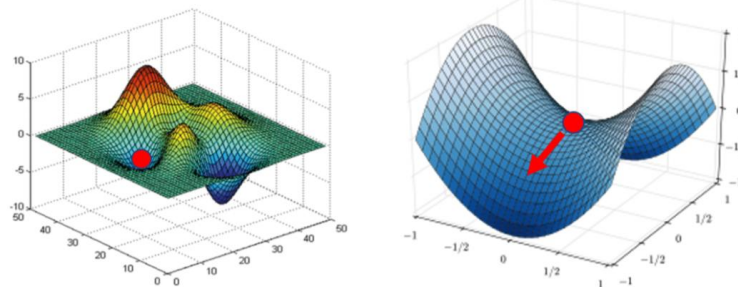
Three intertwined research questions



Methodology

Can medical prior enhance the accuracy and interpretability?

Modality coupling, lesion topology, tumor growth dynamics



Statistical

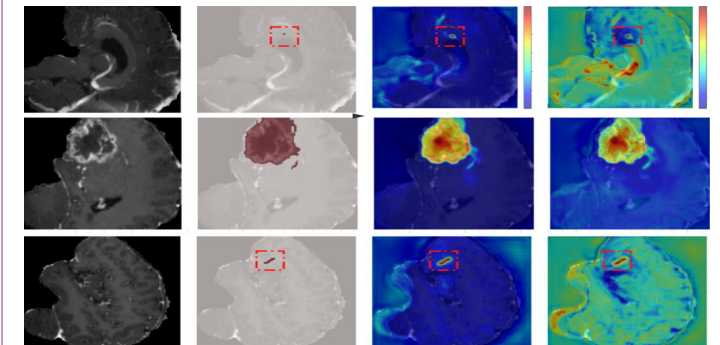
Tversky loss

Focal loss

Which sampling and loss design is best?

weighted sampler

Lesion wise



Interpretive

Does the augmented model behave like a real radiologist?

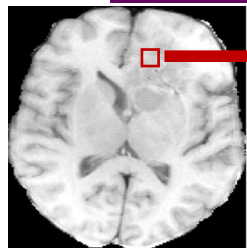
interpretability analysis, visualization

Methodological Direction

Prior enhanced

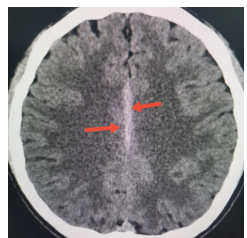
- We have known the **extreme sample ratio** and **the missing modalities** of some data, how to **use these priors to build a new model** that can beat other strong baselines like nnUNet, TransUNet and VM-UMamba?

Sample ratio



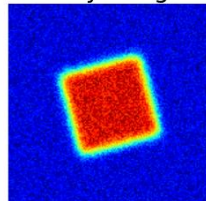
The recurrence is here! So tiny!

Sometimes, the area we are interested in only occupy very small volume of the whole brain



Holes exist in tumor or necrosis

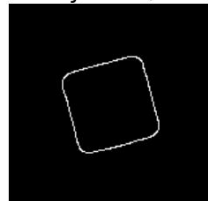
noisy image



Canny filter, $\sigma=1$



Canny filter, $\sigma=3$

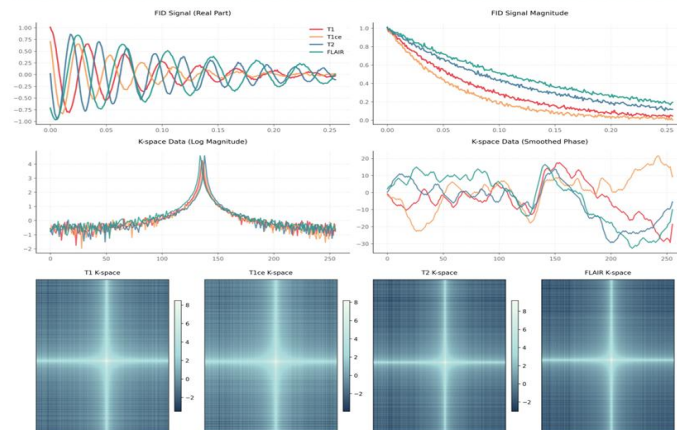


Canny method can help detect the edge and boundry of complex structure

Thus we could use morphology and topology method to help model better understand the brain structure, recurrence and necrosis!

Missing Modalities

There are 4 modalities in brain MRI: T1, T2, T1ce, FLAIR. But some cases only have 1-3 modalities. The modalities we get depend on the MRI machine we set



“A T1ce-only protocol suffices for the segmentation of BMs. The combination of T1ce and FLAIR is important for edema segmentation. Missing either T1ce or FLAIR decreases performance.”

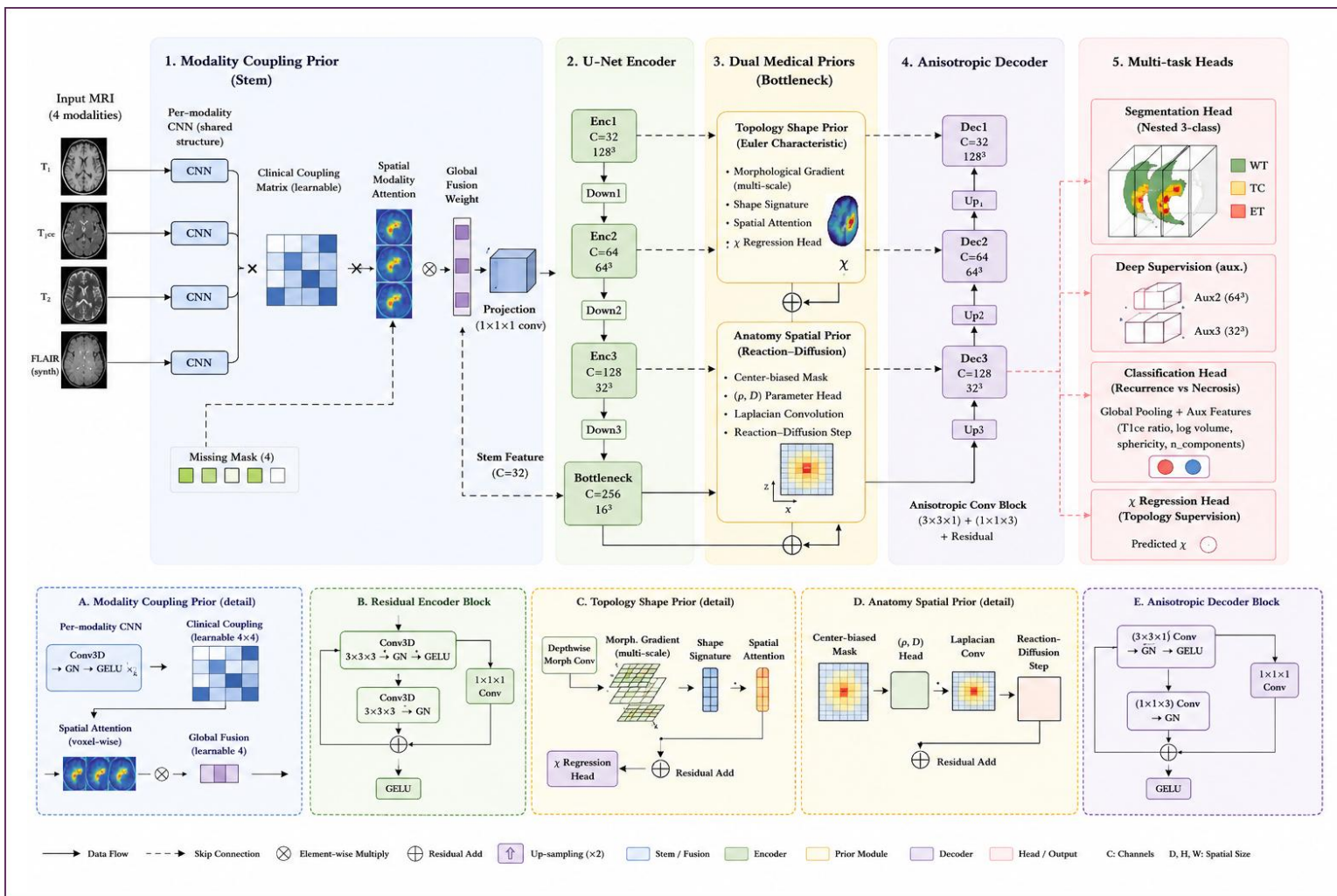
from National Institutes of Health(NIH)

doi:10.1016/j.radonc.2023.109901. Epub 2023 Sep 7. PMID: 37678623

- T1ce and FLAIR are the most important, T1 and T2 are less
- We could *synthesize* better modalities using existing ones!

- Preliminary hypothesis: Injecting medical priors modeled accordingly to domain task DOES WORK WELL

Our Method



BrainTTNet turns EDA-derived medical priors into a lightweight multi-task model for interpretable MRI decision support.

Missing-aware Modality Coupling

- Separate stems for T1 / T1ce / T2 / FLAIR
- Missing-mask gated fusion
- Attention highlights diagnostic modalities

Topology + Anatomy Medical Priors

- Euler χ for solid-vs-cavitory morphology
- Spatial attention over lesion regions
- Bottleneck regularized by medical priors

Multi-task Decision Support Output

- Nested WT / TC / ET segmentation
- Recurrence-vs-necrosis probability
- χ and modality attention for auditability

CONTENTS

- 1 Motivation and Values
- 2 Data Acquisition and Processing
- 3 Research Questions and Methods
- 4 Experiments and Results**
- 5 Team and Roles

Comparision Study and Baselines

Accuracy metrics: Dice, AUC, F1-Score, HD 95, IoU

Efficiency metrics: Flops, Parameters, Training and inferencing time

Main Reasult

Model	Family	AUC	Sens	Spec	Dice	Params
Ours	Prior-aware	0.890	0.830	0.96	0.812	0.15M
Swin-UNETR	Transformer	0.842	0.640	0.95	0.798	62.2M
nnU-Net	U-Net	0.838	0.710	0.95	0.821	31.2M
ResNet10	CNN	0.823	0.778	0.96	-	5.2M
MResNet	CNN	0.850	0.560	0.96	-	8.4M
U-Net	U-Net	0.750	0.500	0.96	0.722	7.5M

Strong Baselines:

CNN family

VGG11 VGG16 ResNet10
ResNet50 DenseNet121
MResNet

U-Net family

U-Net nnU-Net

Transformer family

Swin-UNETR TransUNet
Vision Mamba MedSAM

Key Findings

AUC 0.890:
best overall
discrimination

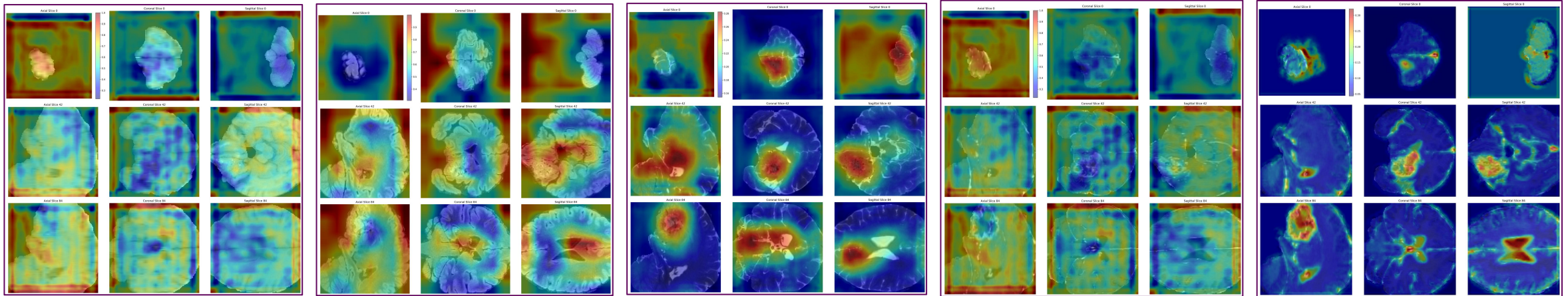
Sens 0.83:
stronger necrosis
detection

<1M params:
lightweight prior-
aware design

Ablation and Interpretation Analysis

Score-CAM Visualization of the decoder layer outputs of different models.

The areas where our model focuses its attention during decision-making closely align with the actual foreground regions, much like how **REAL clinicians** make decisions



Swin-UNETR

nnU-Net

ResNet10

MResNet

Ours

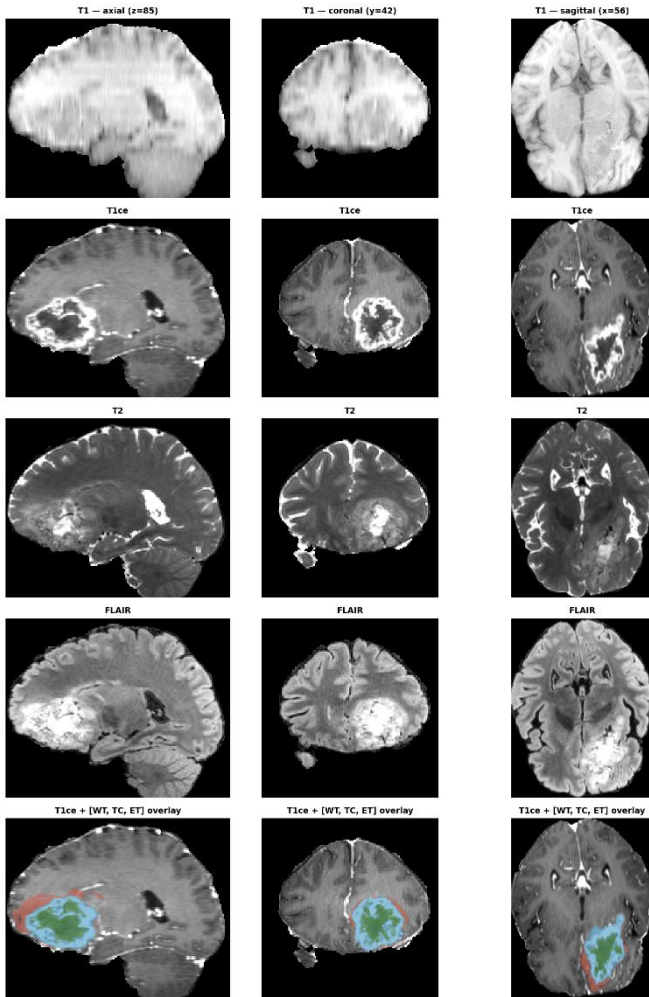
Ablation Study

- ❑ Each modules matters
- ❑ **ModalityCouplingPrior** is the largest contributor, **χ regulariser** is the highest-leverage on sens-on-necrosis at near-zero compute cost

Ablation	Δ AUC vs full BrainTTNet	
Drop ModalityCouplingPrior (concat 1×1×1 conv stem)	-0.048	
Drop TopologyShapePrior (χ -loss off)	-0.040	
Drop AnatomySpatialPrior	-0.029	
All three priors off (plain U-Net)	-0.097	
Zero-fill instead of synthesise FLAIR	-0.014	
BraTS-2021 backbone pretrain	+0.008	
MC-dropout (T = 30 samples) at inference	n/a (CI metric)	

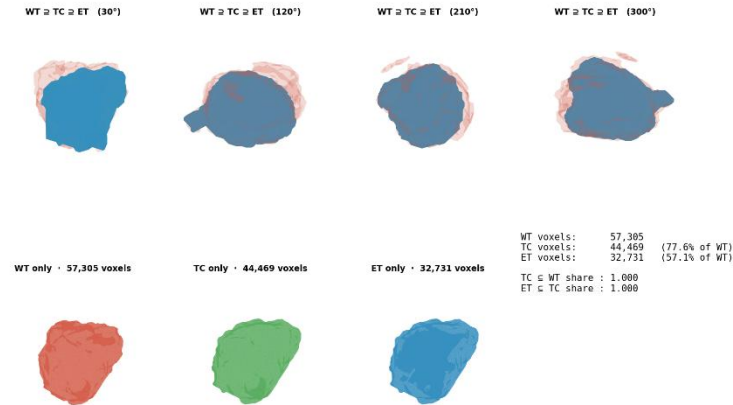
Other Visualization

Case 001 — three-orthogonal-view anatomy panel



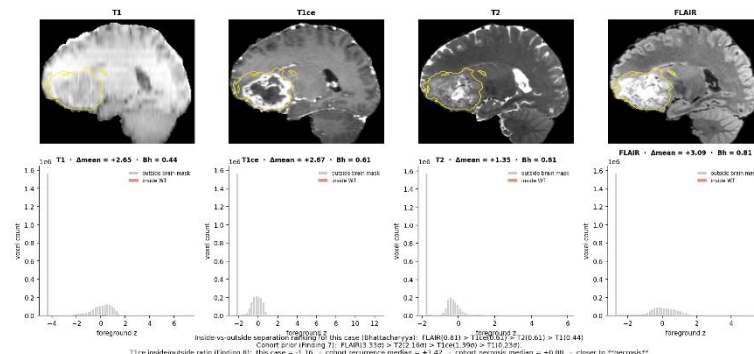
Multimodal anatomy

Case 001 — 3-D nested tumour surfaces (Finding 6 prior)



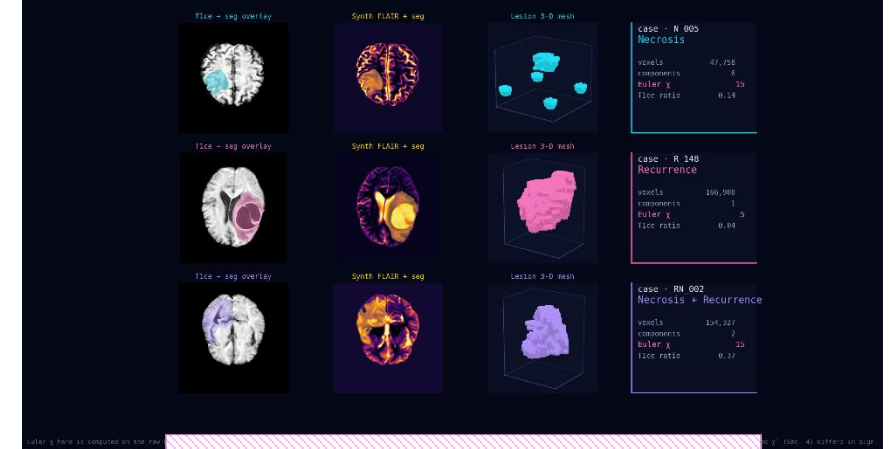
Nested lesion surfaces

Case 001 — modality signature



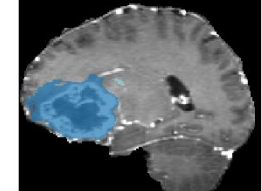
Modality signature

Per-class signatures • how necrosis, recurrence, and mixed lesions look

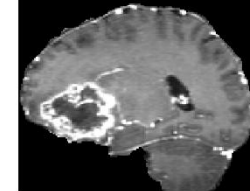


Class signatures

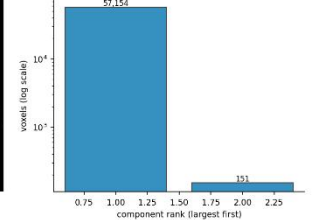
(a) 2 connected components — axial z=85



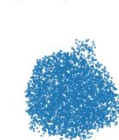
(b) no internal cavities in this slice



(c) component voxel counts · largest = 57,154



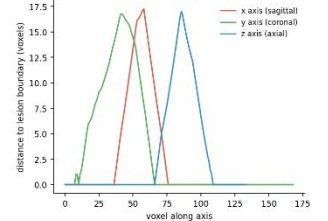
(d) component spatial distribution



```
n components           : 2
largest share          : 0.997
Euler  $\chi$ (WT)          : -1
surrogate holes  $\beta_1$  : 1
cavity voxels          : 3

class signature (cohort priors):
 $\chi \geq -20$   → cavitated necrosis
 $\chi \geq +5$    → compact recurrence
```

(f) distance-to-boundary through centroid



Topology signature

Visualization makes BrainTTNet's medical priors auditable at the single-case level.

Web Design

The website connects clinical motivation, missing-modality synthesis, model performance, and interpretability in one **interactive demo**. <https://braintt.vercel.app>

End-to-End
Model Story

Model Results



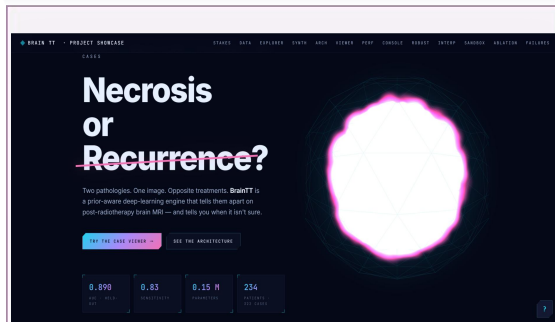
Interactive Evidence



Auditable Review

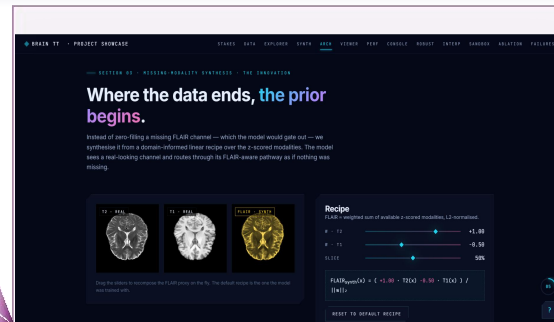
Auditable Decision
Support

Users can inspect cases, test missing-modality synthesis, compare baselines, and audit model explanations.



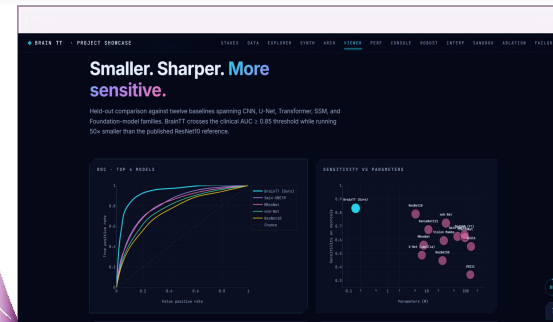
Clinical Problem at a Glance

Presents the R-vs-N task, core cohort, and headline metrics.



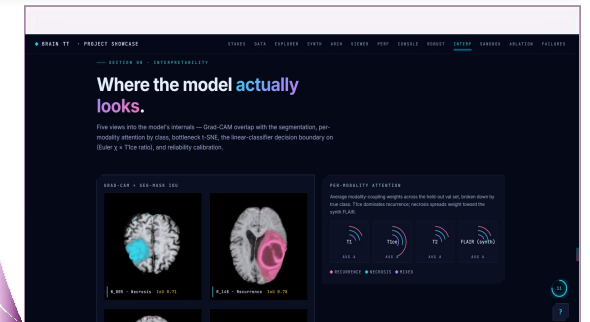
Missing-Modality Synthesis

Interactive FLAIR synthesis from available T1 / T2 signals.



Performance and Baselines

Compares BrainTT with CNN, U-Net, Transformer, and foundation baselines.



Interpretability and Audibility

Visualizes Grad-CAM, modality attention, decision boundary, and calibration.

CONTENTS

- 1 Motivation and Values
- 2 Data Acquisition and Description
- 3 Research Questions and Methods
- 4 Team and Roles
- 5 Timeline and Milestones

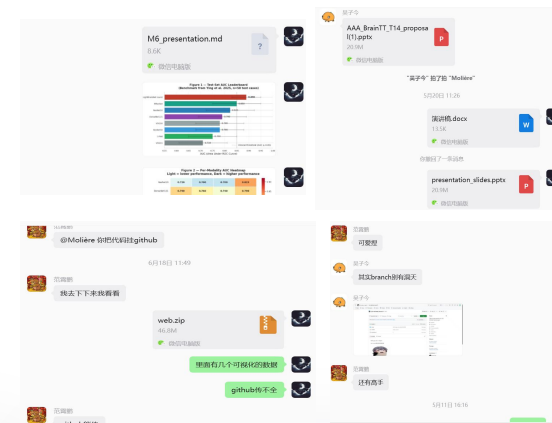
Team and Roles

Leader: Yutong Wang

Principal Researcher
Modeling, network design data
processing, documentation

We collaborate through
Wechat Group, following
Agile developing

Each sprint leader make
plans and deliver tasks



Zijin Wu
ZijinWu1

Zijin Wu

ML Engineer

Network design,
training and
testing



baigei
24baigei

Xiaopeng Fan

Data Scientist

Data curation,
visualization,
conduct EDA

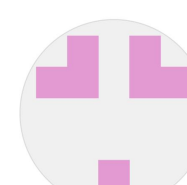


ye52
1234asddf

Ye Wang

Research
Scientist

Evaluation
protocol,
baseline testing



shangyf0528

Yunfei Shang

Product
Manager

Documentation,
visualization,
testing

We have completed research experience and solid research foundation

DIR-Net: A Disentangled Implicit Routing Network For Effective Time Series Forecasting

Jiaxue Yang^a, Yutong Wang^{a,1}, Zhongfeng Kang^{a,*}, Shaoyi Zong^a, Shantian Yang^b, Qinghua Zhao^c and Zichen Song^d

^a School of Information Science and Engineering, Lanzhou University, Lanzhou, 730000, China

^b School of Computing and Artificial Intelligence, Southwestern University of Finance and Economics, Chengdu, 611130, China

^c School of Artificial Intelligence and Big Data, Hefei University, Hefei, 230601, China

^d Department of Applied Artificial Intelligence, Sanggyunkwan University, Seoul, 16419, Korea

ARTICLE INFO

Keywords:
Time Series Forecasting
Temporal Product Attention
Seasonal Pattern Capturing
Adaptive Feature Gating
Kolmogorov-Arnold Network

ABSTRACT

Time series forecasting is pivotal across diverse real-world applications, including energy management, financial trading, and climate monitoring. However, prevailing forecasting architectures often homogenize the processing of trend and seasonal components or rely on predefined fusion strategies. This limits their capacity to effectively capture the heterogeneous temporal dynamics inherent in complex time series. To address these limitations, we propose the Disentangled Implicit Routing Network (DIR-Net), a novel framework that leverages disentangled dual-pathway modeling coupled with an adaptive feature routing mechanism. DIR-Net introduces a Disentangled Temporal Encoder

MSK-Net: Multi-Scale Spatial KANs Enhanced U-Shaped Network For Explainable 3D Brain Tumor Segmentation

Yutong Wang^a, Zhongfeng Kang^{a,*}, Xiaopeng Fan^a, Zijin Wu^a, Shantian Yang^b, Qinghua Zhao^c and Zichen Song^d

^a School of Information Science and Engineering, Lanzhou University, Lanzhou, 730000, China

^b School of Computing and Artificial Intelligence, Southwestern University of Finance and Economics, Chengdu, 611130, China

^c School of Artificial Intelligence and Big Data, Hefei University, Hefei, 230601, China

^d Department of Applied Artificial Intelligence, Sanggyunkwan University, Seoul, 16419, Korea

ARTICLE INFO

Keywords:
3D Medical Image Segmentation
Kolmogorov-Arnold Networks
Explainability
Multi-Scale Feature Fusion
Brain Tumor Segmentation

ABSTRACT

Gliomas, the most common and aggressive primary brain tumors, exhibit substantial morphological complexity and tissue heterogeneity. Accurate segmentation from Magnetic Resonance Imaging (MRI) is critical for diagnosis, treatment planning, and prognosis. Despite recent advances in automated approaches, challenges persist in jointly achieving high 3D segmentation accuracy, computational efficiency, and interpretability. To address these, this study proposes MSK-Net (Multi-Scale KAN-U-Net), a novel 3D segmentation framework that incorporates the Kolmogorov-Arnold Network

Think Globally, Act Locally: A Hierarchical Multi-Agent Framework with Retrieval-Augmented Generation

Yutong Wang^a, Yunfei Shang^a, Zijin Wu^a, Zirui Liu^a, Yina Cao^b and Zhongfeng Kang^{a,*}

^a School of Information Science and Engineering, Lanzhou University, Lanzhou, 730000, China

^b School of Management, Lanzhou University, Lanzhou, 730000, China

ARTICLE INFO

Keywords:
Multi-Agent Systems
Large Language Models
Retrieval-Augmented Generation
Hierarchical Coordination
Information Fusion

ABSTRACT

LLM-powered multi-agent systems have shown strong potential for complex task solving, yet existing flat architectures suffer from quadratic communication overhead, knowledge fragmentation across agents, and an inherent tension between centralized consistency and decentralized efficiency, all of which limit their scalability. We propose CMHA-RAG, a hierarchical multi-agent framework that *thinks globally while acting locally*. The framework couples four components: a three-tier Hierarchical Agent Topology Network (HATN) for structured task coordination, a Hierarchical RAG (H-RAG) mechanism with cascading retrieval and upward knowledge distillation, a Semantic Compression Communication Protocol (SCCP) for efficient inter-agent messaging, and an Adaptive Task Decomposition and Load Balancing (ATDLB) strategy for capability-aware resource

KAN-Boosted Chinese Online Abuse Detection Framework with Sentiment and Toxicity Fusion through Global-Local-Differential Attention

Yutong Wang^a, Zhongfeng Kang^{a,*}, Jiaxue Yang^a, Xiaopeng Fan^a, Zijin Wu^a, Shantian Yang^b, Qinghua Zhao^c and Zichen Song^d

^a School of Information Science and Engineering, Lanzhou University, Lanzhou, 730000, China

^b School of Computing and Artificial Intelligence, Southwestern University of Finance and Economics, Chengdu, 611130, China

^c School of Artificial Intelligence and Big Data, Hefei University, Hefei, 230601, China

^d Department of Applied Artificial Intelligence, Sanggyunkwan University, Seoul, 16419, Korea

ARTICLE INFO

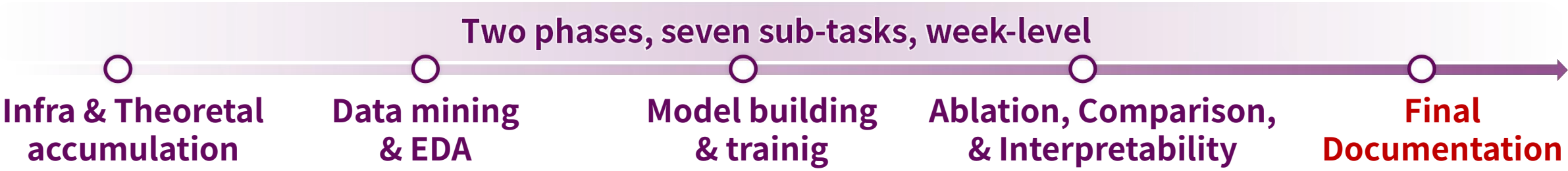
Keywords:
Online Abuse Detection
Sentiment Analysis
Global-Local-Differential Attention
Kolmogorov-Arnold Networks
Feature Fusion

ABSTRACT

Detecting online abuse on digital platforms is a critical yet challenging task, particularly in Chinese social media, where colloquial expressions, emojis, and abstract semantics complicate automated identification. Existing large language models (e.g., OpenAI's GPT) often exhibit limited effectiveness in Chinese contexts, yielding low precision in nuanced scenarios, while smaller task-specific models struggle to capture subtle emotional cues and implicit toxicity. Moreover, most existing architectures emphasize toxicity alone, overlooking sentiment features and thereby constraining overall performance. To address these challenges, we propose Sentox-GLDA, a KAN-boosted framework for Chinese online abuse detection that fuses sentiment and toxicity features through



Gantt Chart



Tasks Decomposition		May. 2026				Jun. 2026				Jul. 2026
Task name		1	2	3	4	1	2	3	4	1
Phase 1	Data curation, EDA, prior design									
	Baseline pipeline, training infra									
	First prior augmented varient									
Phase 2	Prior module ablation study									
	Class imbalance treatment study									
	Interpretability analysis									
	Final documentation									

Limitations and Future Works

Core limitation: scale of the held-out validation split, only 64 cases

- The training cohort is single-institutional because it only contains patients data from Tiantan hospital. When conducting zero shot inference on Huashan data, the AUC drops.

Future Academic Direction

More strong baselines

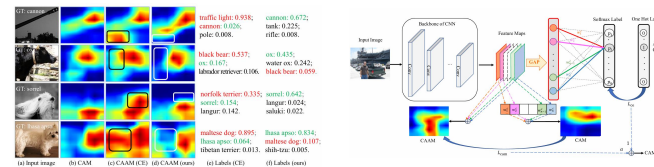
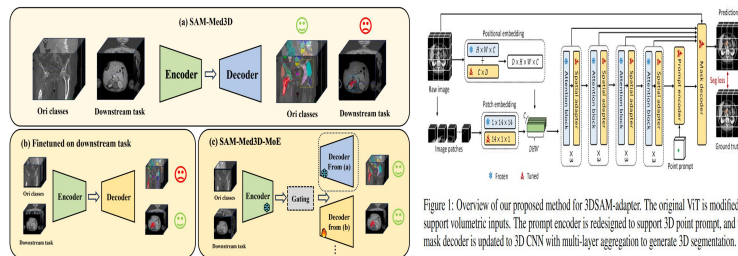
Mamba based models, RWKV based models, SAM based models……

Method Improvement

CAM based loss for more interpretability, tversky loss for imbalanced classes

Academic Paper

Complete the writing of academic paper and submit it during Sep. and Oct.



$$T(\alpha, \beta) = \frac{\sum_{i=1}^N p_{0i} g_{0i}}{\sum_{i=1}^N p_{0i} g_{0i} + \alpha \sum_{i=1}^N p_{0i} g_{1i} + \beta \sum_{i=1}^N p_{1i} g_{0i}}$$

more robust and strong method



MICCAI, IPMI, ISBI, MIDL……

enrich experiments and visualiztion



Thanks for listening

<https://github.com/Emolier-s-hajimi/LZU-INFO442-BrainTT>.

<https://braintt.vercel.app>

Team 14

Leader: Yutong Wang (ID: 320230942461)

Member: Zijin Wu(ID:320230942571), Xiaopeng Fan(ID:320230941801),
Ye Wang(ID:320230942541), Yunfei Shang(ID:320230942271)